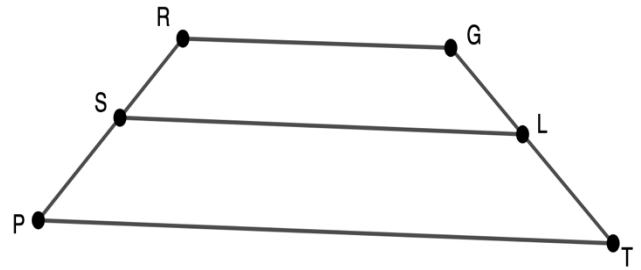


Geometry
Unit 8
Lesson 7 Practice

Name **Answer Key** _____
Date _____
Period _____

$PTGR$ is a trapezoid as shown where $\overline{PT} \parallel \overline{RG}$. $\overline{SL}$ is the midsegment of $PTGR$ with $SL = 18$, $PT = 9x - 5$, and $RG = 7x - 3$.



1. Write an equation to show the relationship between $\overline{SL}$ and the bases of the trapezoid.

$$SL = \frac{RG + PT}{2}$$

2. What is the value of x ? Round your answer to the nearest hundredth.

$$18 = \frac{7x - 3 + 9x - 5}{2}$$

$$36 = 7x - 3 + 9x - 5$$

$$44 = 16x$$

$$x = \frac{11}{4} = 3.75$$

3. What are the lengths of $\overline{PT}$ and $\overline{RG}$? Round your answers to the nearest hundredth.

$$PT = 9\left(\frac{11}{4}\right) - 5 = 19.75$$

$$RG = 7\left(\frac{11}{4}\right) - 3 = 16.25$$

Trapezoid $HKMV$ is shown with midsegment $\overline{DP}$ where $VH = 9x + 12$, $MK = 6x$, and $DP = 10x$.



4. What is the value of x ? Round your answer to the nearest tenth.

$$10x = \frac{9x + 12 + 6x}{2}$$

$$20x = 9x + 12 + 6x$$

$$5x = 12$$

$$x = 2.4$$

5. What is the length of $\overline{DP}$?

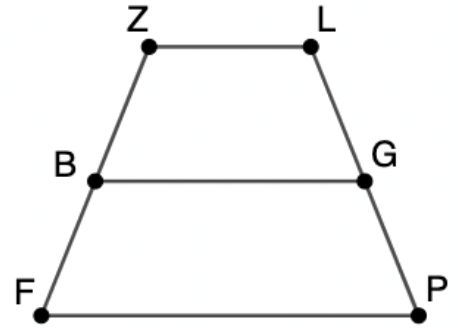
$$DP = 10(2.4) = 24$$

6. What are the lengths of $\overline{VH}$ and $\overline{MK}$? Round your answer to the nearest tenth.

$$VH = 9(2.4) + 12 = 33.6$$

$$MK = 6(2.4) = 14.4$$

Isosceles trapezoid $FPLZ$ is shown with midsegment $\overline{BG}$ where $ZL = 5x + 1$, $FP = 8x + 2$, $BG = 4x + 4$, $BZ = 3.5$.



7. What is the value of x ?

$$4x + 4 = \frac{5x + 1 + 8x + 2}{2}$$

$$8x + 8 = 5x + 1 + 8x + 2$$

$$5x = 5$$

$$x = 1$$

8. What is the length of $\overline{BG}$?

$$BG = 4(1) + 4 = 8$$

9. What are the lengths of $\overline{ZL}$ and $\overline{FP}$?

$$ZL = 5(1) + 1 = 6$$

$$FP = 8(1) + 2 = 10$$

10. What is the perimeter of $FPLZ$?

$$\text{Perimeter} = ZL + LP + FP + FZ$$

$$\text{Perimeter} = 6 + 2(3.5) + 10 + 2(3.5)$$

$$\text{Perimeter} = 30$$